

MCV 4U - Chapter 7: Applications of Vectors Test PM

Name _____

Date May 29, 2025

Total
29.5
/30

1. If $\vec{a} \times \vec{b} = (8, 1)$, then what is $\vec{b} \times \vec{a}$?

a. (8, 1)

b. (-8, 1)

c. (8, -1)

d. (-8, -1)

2. Two forces of 4N and 5N act on an object and their corresponding force vectors have a dot product of 8. What is the angle between the two vectors?

a. 66.42°

b. 27.27°

c. 23.58°

d. 62.73°

3. Given that the vectors $\vec{a} = (6, -2, 2s)$ and $\vec{b} = (-1, s+1, 2)$ are perpendicular, what is the value of s ?

a. -4

b. 0

c. -1

d. 4

$$= (6)(-1) + (-2)(s+1) + (2)(2s)$$

$$= -6 - 2s - 2 + 4s$$

4. Suppose $\vec{p}$ and $\vec{q}$ are vectors such that $\vec{p} \times \vec{q} = (2, -3, 1)$. What is $\vec{q} \times -2\vec{p}$?

a. (4, -6, -2)

b. (2, -6, 4)

c. (-4, 6, -2)

d. (-2, -6, 4)

5. Given $\vec{u} = 3\vec{i} + 6\vec{j} - 2\vec{k}$ and $\vec{v} = (-2, 5, 4)$:

a. Determine the angle that $\vec{v}$ makes with the positive z-axis.

$$\cos \theta = \frac{4}{\sqrt{(-2)^2 + 5^2 + 4^2}}$$

$$\cos^{-1} \left(\frac{4}{\sqrt{2^2 + 5^2 + 4^2}} \right) \Rightarrow \theta = 53.3^\circ$$

b. Determine the vector projection of $\vec{v}$ onto $\vec{u}$. Simplify fully.

$$= \frac{\vec{v} \cdot \vec{u}}{|\vec{u}|^2} (\vec{u})$$

$$= \frac{(3)(-2) + (6)(5) + (-2)(4)}{\sqrt{3^2 + 6^2 + (-2)^2}} (3, 6, -2)$$

c. Determine $-\vec{u} \times 2\vec{v}$

$$-\vec{u} = (-3, -6, 2)$$

$$2\vec{v} = (-4, 10, 8)$$

$$\begin{array}{r} -6 \times 10 \\ 2 \times 8 \\ -3 \times -4 \\ -6 \times 10 \end{array}$$

$$x = -48 - 20 = -68$$

$$y = -8 + 3(8) = 16$$

$$z = -30 + 6(4) = -54$$

$$(-68, 16, -54)$$

6. Find the magnitude of a vector perpendicular to both $\vec{x} = (4, 6, -2)$ and $\vec{y} = (1, 8, 2)$ without using cross product.

let vector = $\vec{a} = (x, y, z)$

$$\vec{a} \cdot \vec{x} = 0$$

$$(x, y, z) \cdot (4, 6, -2) = 0$$

$$4x + 6y - 2z = 0$$

$$\vec{a} \cdot \vec{y} = 0$$

$$(x, y, z) \cdot (1, 8, 2) = 0$$

$$x + 8y + 2z = 0$$

$$\begin{aligned} 4x + 6y - 2z &= 0 \\ x + 8y + 2z &= 0 \\ \hline 5x &= -14y \\ x &= -\frac{14}{5}y \end{aligned}$$

$$4\left(-\frac{14}{5}y\right) + 6y - 2z = 0$$

$$-2z = -\frac{26}{5}y$$

$$z = \frac{13}{5}y$$

let $y = 1$

vector: $\left(-\frac{14}{5}, 1, \frac{13}{5}\right)$

7. Determine the area of the parallelogram determined by the vectors $\vec{a} = (-2, 4, 7)$ and $\vec{b} = (3, 2, -5)$.

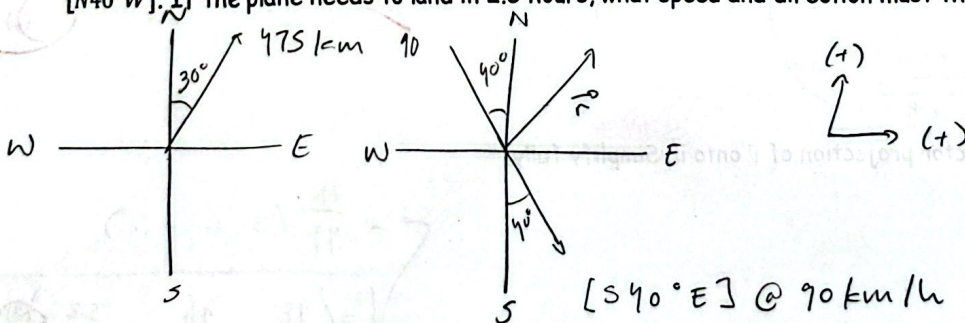
$$a = |\vec{a} \times \vec{b}|$$

$$\begin{array}{rcl} 4 \times 2 & x = -39 \\ 7 \times -5 & y = 11 \\ -2 \times 3 & z = -16 \\ 4 \times 2 & (-39, 11, -16) \end{array}$$

$$= \sqrt{(-39)^2 + (11)^2 + (-16)^2}$$

$$= 39.15 \text{ units}^2$$

8. An airport is 475 km $[N30^\circ E]$ from the current position of an airplane. There is a wind of 90 km/h coming from $[N40^\circ W]$. If the plane needs to land in 2.5 hours, what speed and direction must the pilot fly?



$$d = vt$$

$$v = \frac{d}{t}$$

$$v = \frac{475 \text{ km/h}}{2.5 \text{ h}}$$

$$[N30^\circ E]$$

$$v = 190 \text{ km/h} \rightarrow \text{resultant } \vec{v}$$

$$\text{speed} + \text{direction} + 90 \text{ km/h } [S40^\circ E] = 190 \text{ km/h } [N30^\circ E]$$

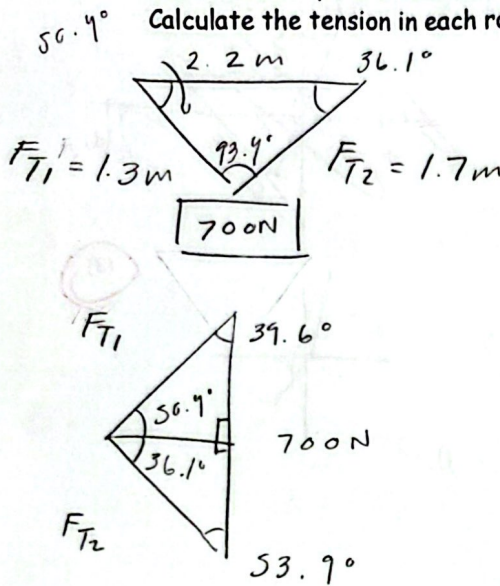
$$190 \text{ km/h } [N30^\circ E] - 90 \text{ km/h } [S40^\circ E] = \text{speed} + \text{dir.}$$

done on other scrap.

$$236.42 \text{ km/h } [E81^\circ N]$$

8.5

9. An object weighing 700N is suspended from two ropes that are attached to a ceiling 2.2m apart. The other end of each rope is attached to a point on the top of the object. The ropes are 1.3m and 1.7 m in length. Calculate the tension in each rope. Include a vector diagram.



$$2.2^2 = 1.7^2 + 1.3^2 - 2(1.7)(1.3)\cos\theta$$

$$\cos\theta = 93.4^\circ$$

$$1.7^2 = 2.2^2 + 1.3^2 - 2(2.2)(1.3)\cos\theta$$

$$\cos\theta = 50.5^\circ$$

$$180 - 93.4 - 50.5 = 36.1^\circ$$

$$\frac{\sin(50.4 + 36.1)}{700\text{N}} = \frac{\sin 39.6}{F_{T2}}$$

$$F_{T2} = 447.0\text{N}$$

$$F_{T1} = 556.6\text{N}$$

$$\frac{\sin(50.4 + 36.1)}{700\text{N}} = \frac{\sin 53.9}{F_{T1}}$$

(4)
The tension of the 1.3m rope is 556.6 N and 1.7 is 447.0 N

10. Consider the Rhombus ABCD, (2)

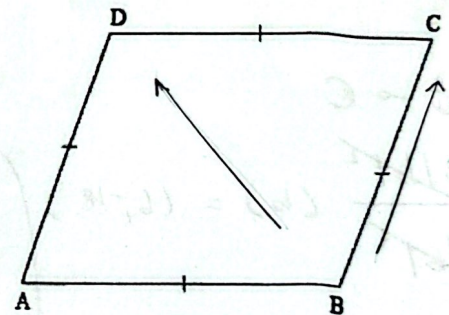
a. State the resultant $\overrightarrow{BC} - \overrightarrow{BD}$

$$= \overrightarrow{BC} + \overrightarrow{DB}$$

$$= \overrightarrow{AB} \quad \checkmark$$

b. State the equilibrant $\overrightarrow{BC} - \overrightarrow{BD}$

$$= \overrightarrow{BA} \quad \checkmark$$



11. Fill in the missing information in the following statement $\vec{a} \times \vec{b} \cdot ? = 0$. Justify your solution. (2)

$$\vec{a} \times \vec{b} \cdot ? = 0 \quad ? = \vec{a} \text{ or } \vec{b}$$

the cross product of $\vec{a} \times \vec{b}$ is a vector that is perp. to both $\vec{a}$ and $\vec{b}$. For a dot product of 2 vectors to $= 0$, the vectors must be perp. $\vec{a}$ and $\vec{b}$ are both perp. to $\vec{a} \times \vec{b}$. Therefore, $|\vec{a} \times \vec{b} \cdot ?| = 0$ if $?$ is $\vec{a}$ or $\vec{b}$.

12. $\vec{b}$ is a vector in $\mathbb{R}^2$. Find the angle between $-\vec{b}$ and $\vec{c} = (2, -3)$ if the following conditions are true.

- Scalar projection of $\vec{a}$ on $\vec{b}$ is 3 times larger than $|\vec{b}|$
- Vector projection of $\vec{a}$ on $\vec{b}$ is $(6, -18)$

vector: $\frac{\vec{b} \cdot \vec{a}}{|\vec{b}|^2} (b) = (6, -18)$ (2)

$$\frac{(x, y) \cdot (2, -3)}{(\sqrt{x^2 + y^2})^2} (x, y) = (6, -18)$$

scalar: $\frac{\vec{b} \cdot \vec{a}}{|\vec{b}|} = 3|\vec{b}| \rightarrow \vec{b} \cdot \vec{a} = 3|\vec{b}|^2$ (1)

$$\frac{(x, y) \cdot (2, -3)}{\sqrt{x^2 + y^2}} = 3\sqrt{x^2 + y^2}$$

$$a \rightarrow e$$

$$\frac{3|\vec{b}|^2}{|\vec{b}|^2} (b) = (6, -18)$$

$$3b = (6, -18)$$

$$3(x, y) = (6, -18)$$

$$(3x, 3y) = (6, -18)$$

$$x = 2, y = -6$$

$$b = (2, -6)$$

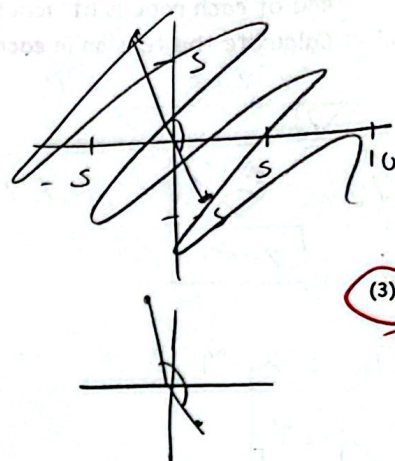
$$-b = (-2, 6)$$

$$\vec{b} \cdot \vec{c} = |\vec{b}| |\vec{c}| \cos \theta$$

$$\cos \theta = \frac{|\vec{b} \cdot \vec{c}|}{|\vec{b}| |\vec{c}|}$$

$$\begin{aligned} \cos \theta &= \frac{(2)(-2) + (6)(-3)}{\sqrt{(-2)^2 + 6^2} \times \sqrt{2^2 + (-3)^2}} \\ &= 169.7^\circ \end{aligned}$$

the angle is 169.7°



(3)

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cont.

Crystal be

$$x \text{ comp: } x = (\sin 30)(190 \text{ km/h}) - (\sin 40)(90 \text{ km/h})$$
$$= 37.149 \text{ km/h [E]}$$

$$y \text{ comp: } y = (\cos 30)(190 \text{ km/h}) - (\cos 40)(-90 \text{ km/h})$$
$$= 233.48 \text{ km/h [N]}$$

$$\text{resultant } \vec{v} = \sqrt{233.48^2 + 37.149^2}$$
$$= 236.42 \text{ km/h}$$

$$\theta = \tan^{-1} \left(\frac{233.48}{37.149} \right) = 81^\circ \text{ [E } 81^\circ \text{ N]}$$

the speed and direction the pilot must fly is @

$$236.42 \text{ km/h [E } 81^\circ \text{ N]}$$